# CCBS 4024 Investments and Portfolio Management



## Topic 4:
## Arbitrage Pricing Theory

# Shortcomings of the CAPM

- One reason for skepticism about the validity of the CAPM is the unrealistic nature of the assumptions needed to derive it.

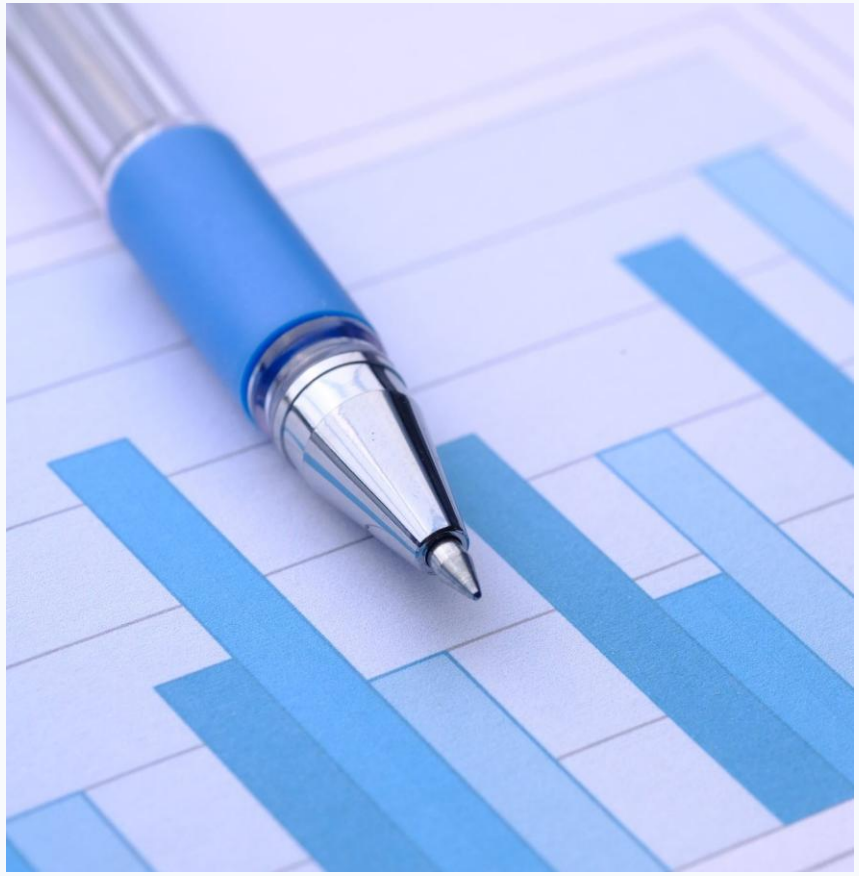
- Some academics proposed to use an alternative approach to asset pricing is called **Arbitrage Pricing Theory (APT)** developed by Ross (1976).

- APT relies on three key propositions:

I. Security returns can be described by a factor model.

II. There are sufficient securities to diversify away idiosyncratic risk.

III. Well-functioning security (Efficient) markets do not allow for the persistence of arbitrage opportunities.



# Single Index Model

- For a single asset say Google, the excess return of Google can be explained by the following statistical model:

$$R_{Gt}^{e} = \alpha_G + \beta_G R_{Mt}^{e} + e_{Gt}$$

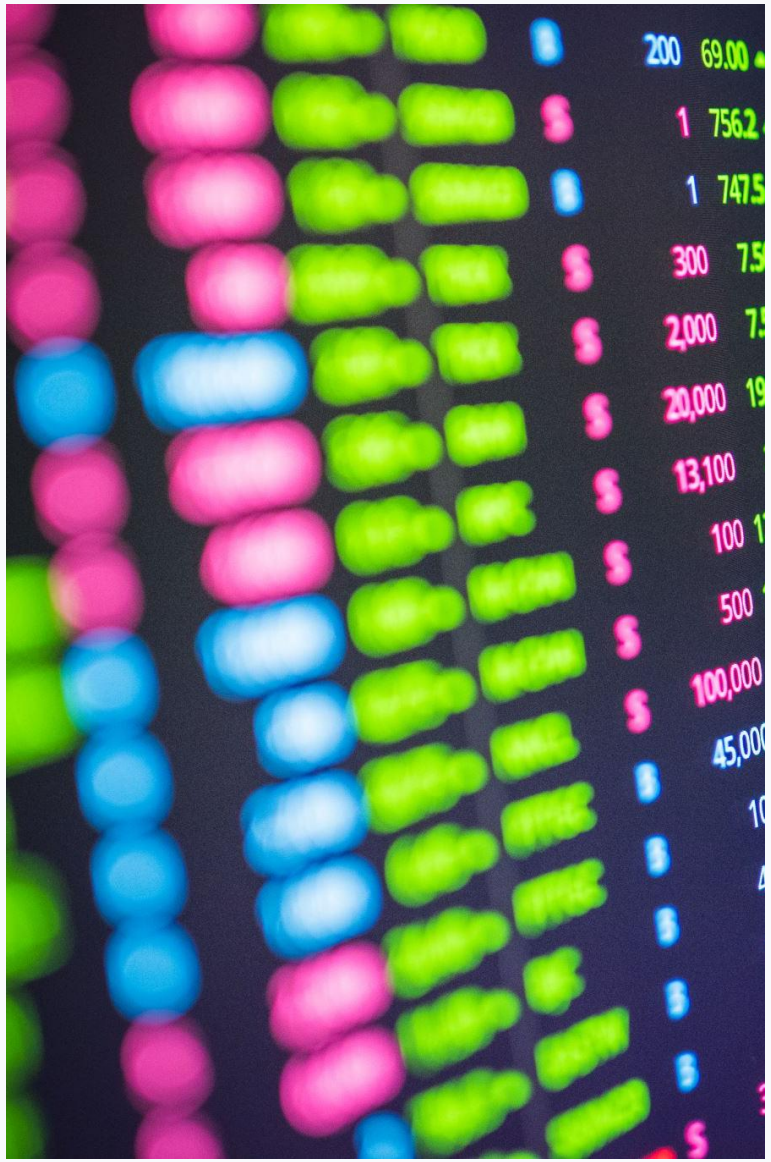
$R_{Gt}^{e}$: Excess return on Google in period t

$\alpha_G$ : Intercept of Google's security characteristic line

$\beta_G$ : Slope of Google's security characteristic line

$R_{Mt}^{e}$: Excess return on market index in period t

$e_{Gt}$: Firm-specific effects (residual term)




# Meaning of Arbitrage

- **Arbitrage** is the act of exploiting mispricing of two or more securities to achieve risk-free profits without making a net investment .
- Suppose shares of a stock sell for different prices on two different exchanges. Then one can make riskless profits by buying the stock with lower price in an exchange and selling the same stock at higher price at another exchange.
- **Arbitrageurs** bid up (down) the price where it is low (high) until the arbitrage opportunity is eliminated or the price of the same asset in two different markets is equalized. This property is called "**Law of One Price**".

# Arbitrage in a well-diversified portfolio

- In a well-diversified portfolio, mean of the residual term is 0. We can drop the residual term from the index model equation as follows:

$$R_p^e = \alpha_p + \beta_p R_{Mt}^e \quad \text{or} \quad r_p - r_f = \alpha_p + \beta_p (r_M - r_f)$$

  where $R_p^e$ is the portfolio's excess return.

- Given a beta value of a portfolio say $\beta_p = 0.7$, we can always form **a mimicking portfolio or replicating portfolio** with the same beta by using the risk-free asset (T-bills) and the market portfolio (the market index).
- By arbitrage-free condition, the well-diversified portfolio and the replicating portfolio should have the same expected returns. That is, there is no Jensen alpha ($\alpha_p$= 0) for the well-diversified portfolio.

# Arbitrage Pricing Theory

- From index model, $r_p - r_f = \alpha_p + \beta_p(r_M - r_f)$.
- Arbitrage-free condition: $\alpha_p = 0$.
- Combining the above, $r_p - r_f = \beta_p(r_M - r_f)$.
- Taking the expected value of the above equation,

$$E(r_p) - r_f = \beta_p(E(r_M) - r_f)$$

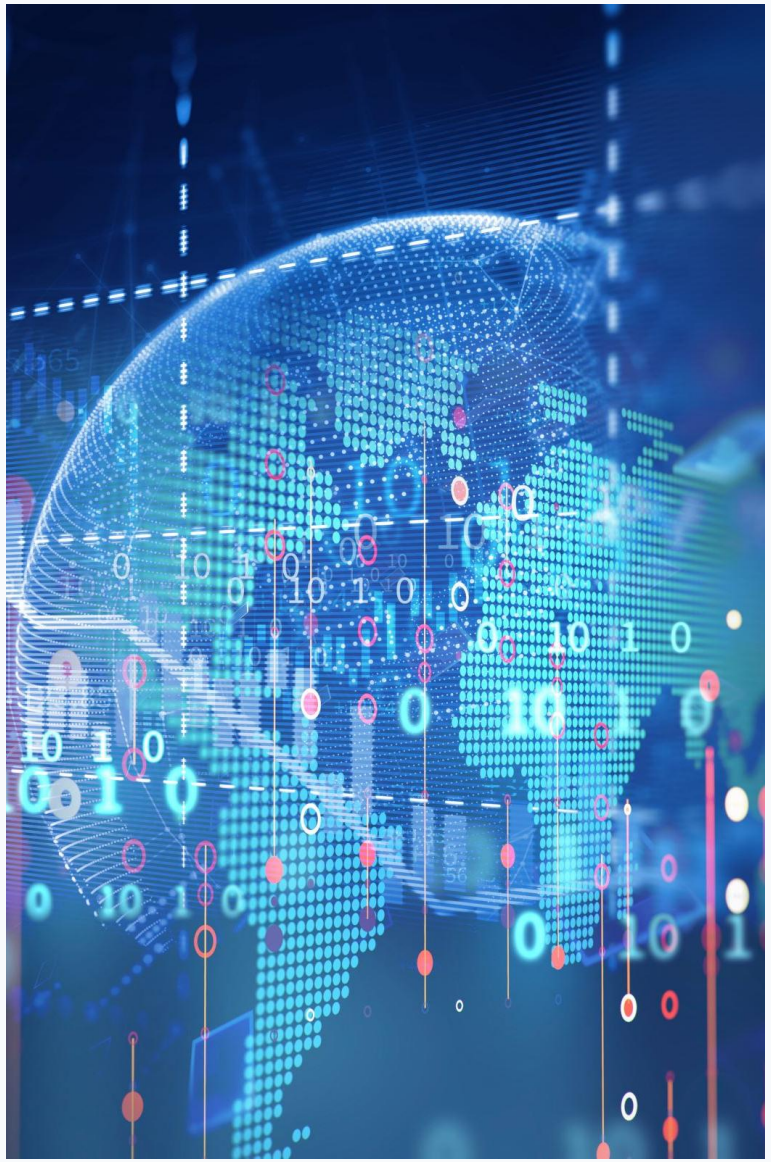
$$E(r_p) = r_f + \beta_p(E(r_M) - r_f)$$

- The above equation is the familiar expected return–beta relation of the CAPM! Its message – that beta and not firm-specific risk should drive asset risk premiums.
- We derived the same CAPM conclusion using the APT with less restrictive assumptions.

# CAPM and APT

- **CAPM**
- Built on an inherently unobservable “market” portfolio.
- Makes a lot of restrictive assumptions such as homogeneous expectations and mean–variance optimizers.
- Pre-assumes the only systematic risk is the market risk. It is a single-factor model.
- Ignores other systematic risk.
- Provides clear statement on the expected return–beta relationship for all securities.

- **APT**
- Built on the foundation of well-diversified portfolios.
- Doesn’t specify the systematic factors that affect securities’ return
- Cannot rule out a violation of the expected return–beta relationship for any particular asset.
- Does not assume investors are mean–variance optimizers.
- Uses an observable market index.
- Can involve more than one systematic risk using multi-factor model



# A Multifactor APT

- So far, we've examined the APT in a one-factor world.
- Actually, there are several sources of systematic risk such as uncertainty in the business cycle, interest rates, energy prices, and so on. Presumably, exposure to any of these factors will affect a stock's appropriate expected return.
- The APT can be generalized to accommodate these multiple sources of risk in a manner much like the multifactor CAPM.

# Two-index model

- Let's start with the two-index model:
- $R^e_{Gt} = \alpha_G + \beta_{M1,G} R^e_{M1,t} + \beta_{M2,G} R^e_{M2,t} + e_{G,t}$
- Assume the excess returns of Google depends on two systematic risk (factors) say the interest rate risk and the market risk called M1 and M2. And Google's sensitivity to these two sources of risk is measured by $\beta_{M1}$ and $\beta_{M2}$ respectively. $\alpha_G$ is the y-intercept of the two-index model and $e_G$ is the firm-specific return of Google.
- Assume investors already hold a well-diversified portfolio, we can ignore $\alpha_G$.
- As there are many assets in the capital market, investors can always form a replicating portfolio that has the same betas as Google.

# Arbitrage and APT

- Thus, Arbitrage-free condition implies $\alpha_G = 0$. And the APT's required expected return on Google is:

$$E(r_G) = r_f + \beta_{M1G}(E(r_{M1}) - r_f) + \beta_{M2G}(E(r_{M2}) - r_f)$$

where

$E(r_{M1}) - r_f$ is called risk premium of Factor Portfolio 1 = expected return on Factor Portfolio 1 – risk-free rate

$E(r_{M2}) - r_f$ is called risk premium of Factor Portfolio 2 = expected return on Factor Portfolio 2 – risk-free rate

# Arbitrage and APT

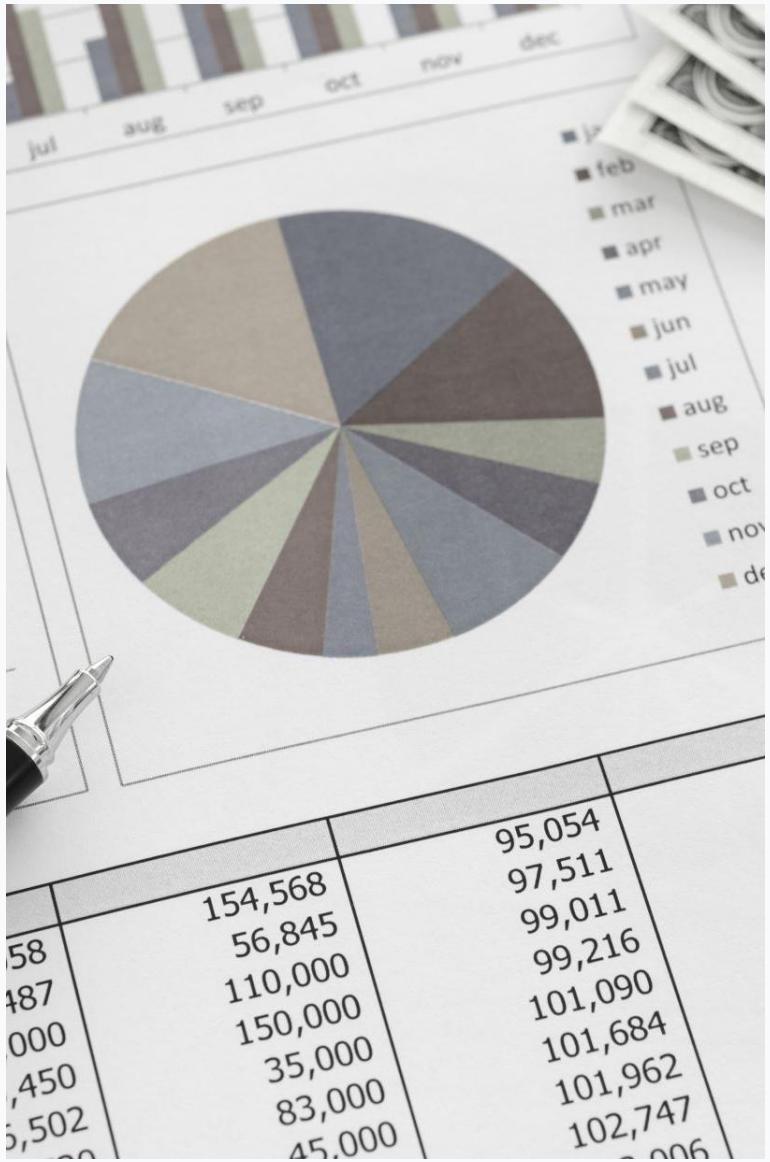
- To generalize, with the APT formula, the expected rate of return of an asset is composed of the risk-free rate plus the product of each risk factor's premium (Expected return of Factor Portfolio i – risk-free rate) and the asset's beta with respect to that factor.

$$E(r_i) = r_f + \beta_{M1,i}(E(r_{M1}) - r_f) + \beta_{M2,i}(E(r_{M2}) - r_f) \cdots\cdots + \beta_{Mk,i}(E(r_{Mk}) - r_f)$$

# Fama-French Three Factor Model

- The Fama-French three-factor model is an asset pricing model that expands on the capital asset pricing model by adding size risk and value risk factors to the market risk factors.
- According to the Fama-French three-factor model, over the long-term, small companies overperform large companies, and value companies beat growth companies. The studies conducted by Fama and French revealed that the model could explain more than 90% of diversified portfolios' returns.






# Question

Assume that two factor portfolios, called Factor Portfolio 1 and 2, carry expected returns of 10% and 12% respectively, while the risk-free rate stands at 4%.
Take a well-diversified portfolio (G) whose beta is 0.5 with respect to the first factor and 0.75 with respect to the second.
Under the Arbitrage Pricing Theory, what equilibrium expected return should Portfolio G offer?